

TABLE OF CONTENTS

M-421 VENTILATION AND AIR CONDITIONING SYSTEM

	Page No.
1. SCOPE OF WORK	M421-2
2. CODES AND STANDARDS.....	M421-3
3. DESIGN AND OPERATING CONDITIONS	M421-4
3.1 General.....	M421-4
3.2 Functions.....	M421-6
3.3 Zone Design Conditions	M421-7
4. DETAILED REQUIREMENTS.....	M421-10
4.1 Centrifugal Fans	M421-10
4.2 Roof Ventilating Fans	M421-11
4.3 Ductwork And Accessories	M421-12
4.4 Airflow Measuring Stations	M421-17
4.5 Terminal Control Units.....	M421-17
4.6 Packaged Air Conditioning Units	M421-18
4.7 Insulation (As Applicable).....	M421-19
4.8 Welding.....	M421-20
4.9 Cleaning And Painting.....	M421-20
4.10 Instrumentation And Controls	M421-20
5. PERFORMANCE TESTING AND GUARANTEES	M421-21
5.1 In Service Testing And Adjustments	M421-21
5.2 Testing, Adjusting And Balancing	M421-21
6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)	M421-23
7. ELECTRICAL AND CONTROL REQUIREMENTS	M421-23

M421 VENTILATION AND AIR CONDITIONING SYSTEM

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Sub Section 1.2 “Summary of Work” and Part 4, Section 3 Technical Specifications S-001 “General Design Requirement of Structure, Civil and Building Works”. Complete Ventilating and Air Conditioning (VAC) system for all structures and buildings shall be furnished by the Bidder to meet all human comfort and electronic and control equipment environmental requirements, including but not limited to the following:

1.1 Steam Turbine Building

1.1.1 Mechanical ventilation by Roof Fans for the Steam Turbine Building

1.1.2 Check the position and area of outdoor Air intake for Ventilation

1.2 Central Control Building

1.2.1 Air conditioning and ventilation system of Central Control Room, Computer and Electronics Equipment Room, Meeting Room, Technical Staff Office, Plant Manager Room, Library Room, Pray Room and other room/area which require personal comfort

1.2.2 Mechanical Ventilation system of LV SWGR Room, MV SWGR Room, Battery Room, Air Conditioning Equipment Room, Pantry, Locker Room, Toilet and Storage

1.3 Cooling Water Pump House

1.3.1 Air conditioning and ventilation system of Electrical Room which require personal comfort and equipment environment requirements

1.3.2 Mechanical Ventilation system of Plant Room

1.4 Water Treatment Plant

1.4.1 Air conditioning and ventilation system of Control Room and Laboratory which required for personal comfort and equipment environment requirements

1.4.2 Mechanical Ventilation system of Plant Room

1.5 Waste Water Treatment Control Building

1.5.1 Air conditioning and ventilation system of Control Room which required for personal comfort and equipment environment requirements

1.5.2 Mechanical Ventilation of Plant Room

1.6 Chlorination Building

1.6.1 Air conditioning and ventilation system of Control Room and Laboratory which require personal comfort and equipment environment requirements

1.6.2 Mechanical Ventilation system of Plant Room

1.7 Emergency Diesel Generator House

1.7.1 Mechanical Ventilation system of Plant Room

1.8 Switchyard Control Building (Existing)

1.8.1 Additional Air Conditioning shall be installed in :

- Control room
- Relay room and
- LVAC room

1.9 Supply the power to whole Air Conditioning and ventilation equipments and required electric equipments

1.10 Support Structures, access galleries, equipment pads, anchor bolts and other auxiliaries as required.

1.11 The systems shall include all necessary plant, equipment, ductwork, pipe work, thermal and acoustic insulation, automatic controls, electronic equipment, cabling and wiring, together with all associated equipment and accessories such as outlet with Weather cover or Vent cap for the efficient working of the systems even if not specified in detail herein.

2. CODES AND STANDARDS

2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

2.2 The codes and standards of the following Organizations extracted from the Part 4, Subsection 2.5 “Codes and Standards” and other equivalent

international Standards are specifically applicable to the design manufacture and testing of the equipment included in this Specification:

AABC	Associated Air Balance Council
ADC	Air Diffusion Council
AFBMA	The Anti-Friction Bearing Manufacturer Association,
AMCA	Air Moving and Conditioning Association
ANSI	American National Standards Institute
API	American Petroleum Institute
ARI	Air-Conditioning and Refrigeration Institute
ASHRAE	American Society of Heating, Refrigerating and Air-
ASME	American Society of Mechanical Engineers
ASNT	American Society for Nondestructive Testing
ASTM	American Society of Testing and Material
AWS	American Welding Society, Inc
HEI	Heat Exchange Institute
IEC	International Electrotechnical Commission
IEEE	Institute of electrical and Electronics Engineers, Inc.
ISA	Instrument Society of America
JIS	Japanese Industrial Standards
NEBB	National Environmental Balancing Bureau
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
OSHA	Occupational Safety and Health Act
SMACNA	Sheet Metal and Air Conditioning Bidders National
UL	Underwriters laboratories, Inc

3. DESIGN AND OPERATING CONDITIONS

3.1 GENERAL

- 3.1.1 The design of the air conditioning and ventilation systems shall be based on the meteorological data specified in Part 4, Sub-Section 1.1.
- 3.1.2 System load and equipment sizing shall be determined by the Bidder, who shall be responsible for the design. Duct Velocity shall not exceed 600m/min.

- 3.1.3 Ventilation systems shall generally be designed to limit the temperature rise in occupied areas to 5° C above the outdoor dry bulb temperature or to limit the temperature in the ventilated areas to maximum 40°C.
- 3.1.4 Air conditioning systems shall generally be designed to limit the temperature and relative humidity to protect electric equipment in accordance with the criteria specified in Sub Section 2.16 Environmental Considerations and to provide comfort for human occupancy. For air-conditioning areas, the indoor temperature shall be no more than 24°C± 2.0 °C with a relative humidity of 50% ± 10%.
- 3.1.5 Air shall be distributed in a manner which ensures freedom from objectionable draughts or rapid changes of temperature. The air velocity in duct shall be such that the pressure drop does not exceed 2.5 mm of water column per 30 m length of ductwork. Under all loading conditions the systems shall be free of intrusive noise or structure-borne vibration.
- 3.1.6 Fresh air intakes shall be located so as not to draw in hot air, dusty air, circulated air or exhaust air and shall be protected against the penetration of rain, birds and foreign matter.
- 3.1.7 Equipment within classified hazardous areas particularly roof extraction and battery room ventilation fans shall be suitable designed to comply within the requirements of API 500A or equivalent.
- 3.1.8 Design and installation shall be generally in accordance with the recommendations of the American Society of Heating, Refrigeration and Air-Conditioning Engineers of equivalent recognized institution and particularly to the requirements of the Indonesian Fire Department.
- 3.1.9 Protective devices and interlocks shall be provided so that equipment in the system will start up operate or shutdown in a proper sequence to ensure a smooth and safe performance of the system.
- 3.1.10 The ventilation system shall provide a minimum of 10 room air changes per hour of uniformly distributed outdoor air.
- 3.1.11 Supply air of the exhaust ventilation system shall be filtered to provide a dust free atmosphere. The air filters shall have a minimum 30 % efficiency, based on the filter grade as F5 to EN779 (ASHRAE STANDARD 52-76)
- 3.1.12 Centrifugal fans shall be provided for the area where require mechanical ventilation system to maintain positive pressure in order to prevent the infiltration of dust from the adjacent areas.
- 3.1.13 The Air Conditioning System shall be condensing unit plus air handling unit type or separate type air conditioner.

3.2 FUNCTIONS

The functions of the ventilation and air conditioning systems are to:

- Protect equipment and personnel from the extremes of temperature.
- Maintain comfort conditions in continuously occupied zones.
- Provide outside air for smoke purging.
- Remove detrimental air-borne contaminants
- Provide temperature, humidity and dust control for sensitive equipment
- Remove hazardous gases.
- Provide fresh outdoor air for personnel occupancy.

3.3 ZONE DESIGN CONDITIONS

3.3.1 Ventilation for Steam Turbine Building

3.3.1.1 Roof fan

A sufficient number of exhaust type Roof fan with shutter moved by motor and bird screen shall be mounted on the roof to remove hot air. The materials of Roof fan shall be appropriate for the seacoast location and subject to review by the Owner.

3.3.1.2 Required Air Quantity for Ventilation

The whole air quantity extracted by total Roof fans shall be satisfied to meet the more critical of the following requirement

- a). Maintain the temperature of the Steam Turbine Building below 40°C for the area under overhead traveling crane rail level when each Generator is operational.
- b). The extract fan for the cable mezzanine of Steam Turbine Building shall be provided two (2) units, and its duty air quantity is more 10 times room air change per hour.

3.3.2 Ventilation and Air Conditioning System for Central Control Building

3.3.2.1 Description of Service of Air Conditioning System.

- a). Air Conditioning System shall be provided with air-conditioning equipment which is consisted in 2 x 100% capacity and connected a ducted air distribution and return system concealed in the ceiling.
- b). High efficiency air filters, such as filter grade F6 to EN779 or equivalent, shall be fitted to this system which required by the electrical equipment.
- c). The duplicate equipment shall be arranged for manual changeover during normal operation and automatic changeover in the case where failure of a.c. electrical supply is followed by near immediate resumption due to unscheduled loss of supplies.

Each of air conditioning equipments which are capable of supplying the total air conditioning load of the zones served when operating in one running, one standby mode.

- a) The inlet outdoor air quantity of air conditioning room shall be provided with a minimum of 30 m³/h per a person living room.

- b) Indoor Air Conditions in the air conditioning room shall be maintained within the following range:

For Central Control Room and Computer and Electronics Equipment Room

Temperature : $24^{\circ}\text{C} \pm 1^{\circ}\text{C}$

Relative Humidity : $45\% \pm 5\%$

For Other Room

Temperature : $24^{\circ}\text{C} \pm 2^{\circ}\text{C}$

Relative Humidity : 50% (Design)

- c) Noise levels due to Air Conditioning Equipment and System noise shall not exceed NC 40 in any conditioned.
- d) The systems shall be of the low pressure, low velocity constant volume type with air volume control by damper for temperature control of the zones specified above.

3.3.2.2 Ventilation for Battery Room

- a) The air quantity of exhaust ventilation system shall provide a minimum of 10 of Battery room air changes per hour of uniformly distributed outdoor air.
- b) If these rooms have some heating equipment, that shall meet the requirements of Clause 3.1.3.
- c) Exhaust fans shall be provided to maintain a slight negative pressure to prevent release of hazardous H_2 gas to surround areas.
- d) Number of units required: 2 x 50% Duty for Battery Room with emergency power required.

3.3.2.3 Ventilation for Toilet

The exhaust air quantity of toilet shall be provided for a minimum with 15 room changes per hour of uniformly distributed outdoor air

3.3.2.4 Ventilation for other rooms

- a) The quantity of air change which is the following rooms shall be provided as follows:

<u>Room Name</u>	<u>Duty Air Change [times per hour]</u>
Storage	6
Pantry and Locker Room	10
Air Conditioning Equipment Room	10

- b) That of the LV and MV SWGR Area shall meet the requirements of Clause 3.1.3.
- c) The ventilation system shall provide a minimum 20 times air changes for the amount of theoretical combustion air when burning appliance will be provided in the room,

3.3.3 Ventilation and Air Conditioning system for other buildings.

- 3.3.3.1 Air conditioning system with ventilation shall be provided in the following room so as to maintain an environment suitable for equipment operation and human occupancy.

<u>Building Name</u>	<u>Room Name</u>
Cooling Water Pump House	Electrical Room
Water Treatment Plant	Control Room, Electrical Room
Water Waste Treatment Control Building	Control Room
Chlorination Building	Control Room
Rest Space, Guest Room	Switchyard control room
Switchyard	(Existing)

And, air Conditioning systems shall meet the requirements of Clause 3.3.2.1.

3.3.3.2 Ventilation System

- a) Description of Services

The following extract fans, but not be limited to, shall be provided:

Table 4.2.1 Preliminary Number of Fans to Install

<u>Room Name</u>	<u>Building</u>	<u>Number of Installed Fans**</u>	<u>Duty (Air change/hr)*</u>
Plant Room	each	2	10
Workshop	Workshop Building	4	10
Mix Bed/Chemical Room	Water Treatment	2	10

Notes :

* Duty as described above is the air handling capacity of total installed fan in terms of equivalent air-changes per hour in the respective ventilated space.

** The number of installed fans shall be finalized subject to detailed design.

3.3.4 Work of Ventilation Fan Installation

Extract fans shall be arranged for wall mounting or roof mounting and shall be complete with direct-driven axial fan selected for peak efficiency, heavy gauge wire mesh guards on inlet and outlet, self-closing air stream-operated weatherproof dampers and external weather cowl and flashing.

3.3.5 Control Requirements of Ventilation system

Each unit shall be controlled from an 'Off/Auto/Run' selector switch. When its selector switch is set in the 'Auto' position, the fan shall be controlled by its own adjustable set-point room thermostat so that the fan runs when the temperature in the ventilated space exceeds the thermostat set point and the fan stops when the temperature in the ventilated space falls below the set point. Thermostats shall be adjustable over a range of at least 15°C to 40°C and shall have a 2°C dead band.

4. DETAILED REQUIREMENTS

4.1 CENTRIFUGAL FANS

The Bidder shall comply with the following specification when a centrifugal fan is provided for the ventilation and air conditioning system.

4.1.1 Centrifugal fans shall be statically and dynamically balanced, and have a high efficiency wheel with backward inclined blades. The drive arrangement for fans shall be belt driven AMCA 1 SWSI with heavy-duty integral isolation

base and have vibration isolators and motor drive belt adjustment. Each fan, motor V-belt drive and belt guard shall be completely assembled at the factory on the integral isolation base.

- 4.1.2 Fans shall be industrial grade continuous duty type. Fan side sheets and scroll sheets shall be of continuous welded construction. Materials of construction shall be manufacturer's standard carbon steel except that materials for rotors, blades, shafts, vanes etc. shall be as appropriate for the location, service and operating conditions.
- 4.1.3 Fan and motor shaft bearings shall be designed for grease lubrication. Fan bearings shall be rated for a minimum L-10 life of 100,000 hours per AFBMA (Anti-Friction Bearing Manufacturer Association).
- 4.1.4 All exposed external rotating parts such as couplings, pulleys, etc. shall have suitable guards conforming to OSHA requirements as a minimum.
- 4.1.5 Each fan shall be provided with lifting eye or suitable lugs designed to permit hoisting the fan completely assembled to its isolation base.
- 4.1.6 Each fan shall have the fan housing scroll furnished with bolted and gasketed clean-out doors. In addition a housing drain with valve shall be provided for each fan at its low point.
- 4.1.7 The fans shall be of the quietest type commercially available and their noise criteria shall conform to the requirements of Part 4, Subsection 2.9. Bidder shall guarantee the octave band sound power levels for each different fan model. Sound power level ratings shall comply with AMCA Bulletin 301, and shall be the result of tests made in accordance With AMCA Bulletin 300.

4.2 ROOF VENTILATING FANS

- 4.2.1 The fans shall be manufacturer's standard propeller fan complete with mushroom housing, motors, limit switches, access doors and terminal boxes.
- 4.2.2 Quick opening and closing, hinged access doors shall be provided in the fan stack for easy side access to the fan motor, dampers, operators, limit switches, belts, pulleys, bearings etc. as may be required for maintenance or removal and replacement of any of the above items. Inspection and access doors shall be of rigid construction provided with suitable gaskets and clamps so that the doors are airtight. Doors shall be labeled for identification and capable of being opened by hand without the use of a tool. To facilitate fan motor removal through the hood, some suitable method shall be furnished such as lifting eye bolts on the interior of the hood.

- 4.2.3 Mushroom hoods shall be designed with ample open area on all four sides of the stack to minimize air velocities and pressure drop. Hoods shall be constructed of galvanized steel sheets with rain-tight seams.
- 4.2.4 Thermostats shall be furnished as required and be designed for temperature range from 20 °C minimum to 40 °C maximum.
- 4.2.5 V-belt fan drive shall be designed for heavy duty high torque applications and for not less than 200% of the maximum motor torque.
- 4.2.6 All fan bearings shall be furnished with a minimum L-10 life of 100,000 hours per AFBMA and with suitable seals for operation in a dusty atmosphere.
- 4.2.7 The Bidder shall provide bird screens which shall be galvanized 20 mm square mesh.
- 4.2.8 The fans shall be of the quietest type commercially available and the noise criteria shall conform to the requirements of Part 4, Subsection 2.9.
- 4.2.9 The fans shall be rated in accordance with AMCA Bulletin 210.

4.3 DUCTWORK AND ACCESSORIES

Ventilation and air conditioning ductwork shall be provided in all areas so as to achieve a uniform distribution of air, to direct air flow over hot spots and to locate air intakes and exhausts in suitable locations.

All supply registers and diffusers shall be sized for the face and neck velocity range as between 2.5 m/sec to 5 m/sec with mature consideration of the install location and size for sufficient effect.

All return registers and grilles shall be sized for a maximum face velocity of 1.5 m/sec.

4.3.1 Materials

- 4.3.1.1 All Ductwork shall be galvanized and in accordance with the following:
Material Coating

Material	Material Type	Size	Material Specification	Coating Designation
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Sheet	Galvanized Steel	All	ASTM A653 or JIS G3302	ASTM A653 G-90
Stiffeners, Hangers and Support	Carbon Steel, Hot Dipped galvanized	All	ASTM A575, Grade M-1020 and A36 or JIS G3101 JIS G3350	ASTM A123 or JIS G 3302, H 8641
Bolts	Galvanized Steel	All	ASTM A307 or JIS B1180 JIS B1181	ASTM A153 Class D, Commercial Coating
Rivets	Galvanized Steel	All	ASTM A31 ASTM A502 or JIS B1213	
Sheet Metal Screws	Galvanized Steel	All	ASTM A548 or JIS B1115	

Notes : Thickness Duct material : 0.5 ; 0.6 ; 0.8 ; 1.0 ; 1.2 ; and 1.6 mm
Galvanized coating for steel : 244 to 281 gram/m² (JIS G 3302)
Nominal galvanized coating : 275 gram/m² (ASTM A653–G90)

4.3.2 Galvanized Steel Duct Constructions

4.3.2.1 Rectangular Ducts

All galvanized steel rectangular ductwork shall be constructed of thicknesses and stiffened as given in the following table :

Duct Dimension, mm	Minimum thickness, mm	Angle Stiffeners or <u>Hangers at point of Support</u> Minimum Size, mm	Minimum Spacing, mm
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Up thru 600	0.76	25 x 25 x 3	1200 o.c
625 to 900	0.90	40 x 40 x 3	900 o.c
Plenums or as Specified	1.90	50 x 50 x 5	600 o.c.

Notes : o.c = on center , Duct should not exceed 1:4 short to long side

- a) Seams and joints:
 - a.1) Longitudinal seams for ducts less than 1.5 mm thick shall be double-locked, groove type. Ducts 1.5 mm thick and heavier shall have welded longitudinal seams. Any seams prepared in the field must meet design and quality standards of shop prepared seams.
 - a.2) Duct 1.2 mm thick and heavier shall have companion angle flanged joints with sealant or gasket. Flanges shall be the same size as angle stiffeners, but not less than 40 x 40 x 3 mm, or the size of the hanger angle, if the hanger is used for the joint flange.
- b) Bidder shall provide stiffeners and reinforcements so that the ducts will be free from excessive deflection, vibration, drumming or noise. Additional stiffeners shall be provided without additional cost if, in the opinion of the Owner, they are required to eliminate excessive deflection, vibration or noise. Angle iron stiffeners shall be 25 mm stitch-welded to the ductwork on not greater than 225 mm centers.
- c) Hangers and Supporting Systems
 - c.1) Rectangular ducts shall be supported on not more than 2.5 m intervals.
 - c.2) Hangers and supports shall be angle iron of size not less than that used for duct stiffening. The use of strap hangers will not be permitted.
 - c.3) Bidder shall furnish and install all hangers, supports and braces as required for the entire duct systems and do all necessary bolting and welding for such hangers and supports. The most suitable location for each hanger or support and the use of available supporting structures shall be determined by the Bidder at the site. Transverse welding across flanges will not be permitted. Any additional structural steel members not in place, which may be necessary to properly hang and support the duct systems, shall be furnished and installed by Bidder.

Location of all such structural steel and method of attachment to existing steel shall be as approved by the Engineer.

- c.4) Ducts shall be supported with hangers to suit the construction in accordance with the SMACNA. (Sheet Metal & Air Conditioning Contractors National Association)

4.3.3 Penetration Seals

- 4.3.3.1 All sealants and gaskets utilized by Bidder shall be suitable for the respective location and environmental exposure to which the sealant materials will be subjected.

- 4.3.3.2 Bidder shall furnish and install galvanized flashing on all insulated vertical risers which penetrate floors.

4.3.4 Flexible Connections

Connections between the ductwork and vibrating equipment, such as fans and air handling units, shall be made with flexible material.

4.3.5 Flexible Ducts

- 4.3.5.1 Flexible ducts shall be used to connect ductwork to terminal control units (mixing boxes, etc.) and shall be of the same diameter as the inlet connections on the units. Flexible ducts to diffusers shall be the same size as the inlet neck.

- 4.3.5.2 Flexible ducts shall meet all applicable requirements of NFPA 90 A.

- 4.3.5.3 Flexible duct shall be installed in accordance with SMACNA Flexible Duct Installation Standard and in accordance with UI listing requirements.

4.3.6 Access Doors

- 4.3.6.1 Bidder shall provide access doors in all duct plenums in all ductwork on the upstream side of each air flow measuring station, mitered elbow, and control damper, and on either side of each humidifier and fire damper and control damper and duct mounted fans.

- 4.3.6.2 Doors shall be located and sized to facilitate inspection and/or cleaning of the duct-installed equipment. The size of door shall be mature considered to provide sufficient maintenance work and less than the height or width of duct.

- 4.3.6.3 All duct mounted access doors shall be sealed by gaskets and hinged and shall be leak tight construction.

4.3.7 Air Diverting Devices

- 4.3.7.1 Bidder shall provide all splitter dampers or volume extractors required to balance each ventilation system. One adjustment rod shall be provided for splitters up to 300 mm high. For splitters greater than 300 mm high,

adjustment rods shall be provided with spacing no more than 300 mm on center.

4.3.7.2 Construction details shall be in accordance with SMACNA Standard.

4.3.8 Balancing Dampers

Bidder shall furnish and install manual balancing dampers as may be required to properly balance each ventilation and air conditioning system. Balancing dampers shall be manually operated.

4.3.9 Multiple-leaf Dampers

4.3.9.1 Damper frames shall be not less than 2.7 mm thick, galvanized steel formed into a channel having not less than a 150 mm web and not less than 60 mm flanges. Damper frames shall be electric arc welded.

4.3.9.2 Damper frames shall be double-skin airfoil type. Width of blades shall be not greater than 250 mm and not less than 150 mm. Blades shall be fabricated from not less than 1.5 mm thick galvanized steel. Each blade shall be of two-piece, galvanized steel, formed into an airfoil shape with one side rolled over at the edges and bolted to the center shaft.

4.3.9.3 Blade shafts shall be not less than 18 mm diameter cold finished bar. The shafts shall extend the entire length of the blade plus the distance beyond the blade through the frame to the linkage at the exterior side of the damper. The blades shall be through bolted to the shaft.

4.3.9.4 The damper linkage shall be outside the air stream and not less than 6 mm x 30 mm flat, cold finished bar. Linkage shall be tight, vibration free and shall permit no independent action of an individual blade.

4.3.9.5 Bearings for all dampers shall be sintered bronze type, press fitted into the frame.

4.3.10 Fire Dampers

4.3.10.1 Bidder shall furnish and install 1.5 hour fire dampers and sleeves at every penetration of any fire compartment floor, wall or partition.

4.3.10.2 Fire dampers shall be solid steel curtain type with interlocking blades, galvanized steel construction and complete with 70°C UL listed fusible link.

4.3.10.3 Fire dampers and sleeves shall be installed in accordance with the Manufacturer's installation instruction.

4.3.11 Grilles: Registers and Diffusers

4.3.11.1 Grilles, registers and diffusers shall be used to distribute air uniformly throughout the conditioned space and shall be selected and sited to ensure uniform distribution of air and temperature differentials no greater than 1 °C.

4.3.11.2 Each type of grille, register and diffuser shall be tested in accordance with the Air Diffusion Council (ADC) Equipment Test Code 1062

4.3.12 Ventilation Filters

4.3.12.1 Bidder shall furnish filters and filters shall be provided at each fresh air inlet as required. Filter and pre-filter support frames shall be factory-fabricated and assembled. All joints between frames and plenums shall be caulked with duct sealant to minimize air bypass.

4.3.12.2 All filter elements shall have a UL Class 2 rating.

4.3.12.3 High efficiency filters shall be self-supporting.

4.3.12.4 Elements during initial installation and pre-operating testing. After system balancing and prior to turning the system over to the Owner, Bidder shall remove the filters and pre-filter elements and shall install only new, clean, high- efficiency filter elements complete with pressure differential alarm across filter.

4.4 AIRFLOW MEASURING STATIONS

4.4.1 Bidder shall furnish and install permanent airflow measuring stations on the main ductwork and all branches.

4.4.2 All air measuring stations installed within five hydraulic diameters of a fan shall be designed specifically for installation in the fan discharge.

4.4.3 Each airflow measuring station used for balancing the system shall measure airflow by total, and static pressure sensors. Each unit shall consist of a network of pivot tubes, factory-positioned and connected in parallel to produce an average velocity pressure, which is indicative of the total airflow. The pivot tubes may be removed after the systems are balanced.

4.5 TERMINAL CONTROL UNITS

4.5.1 The Terminal Control Units (TCU) shall be rated for airflow, pressure drop, leakage and Sound power level in accordance with ADC Test Code 1062

4.5.2 Constant volume units shall be pressure independent at the design airflow. Mechanical volume regulators are unacceptable.

- 4.5.3 Variable volume units shall be pressure independent and shall reset to any airflow between the minimum and maximum. The use of flow limiters is unacceptable.
- 4.5.4 TCU's shall have their controls factory-set to the specified capacities but shall have provision for adjusting capacities in the field.
- 4.5.5 The airflow shall not vary more than five percent from a given setpoint when the upstream static pressure varies from the required minimum to 75 mm. wg.

4.6 PACKAGED AIR CONDITIONING UNITS

- 4.6.1 Packaged air-conditioning units shall be factory assembled complete with insulated hermetically sealed cabinets, compressor, air or water cooled condensers, pressure regulating valves, condenser fans and motors, refrigerant filter and driers, piping and thermostatic expansion valve, sight glass, discharge, suction and liquid line service valves; unit controls, high and low pressure switches, holding charge of refrigerant, compressor crankcase heater, high pressure relief valves, evaporator coil(s), disconnect combination circuit breakers and starters, power and control panels. Each unit shall have built-in contactors, fan relays and thermostat for remote mounting with on-off switch.
- 4.6.2 Fans shall be centrifugal type, AMCA certified in accordance with AMCA Ski 10, complete with electric motor drive with adjustable motor pulley. Fans shall conform to Clause 4.2 herein. Each fan unit shall have belt drive. Each drive shall consist of fan and adjustable pitched motor sheaves and a minimum of two matched and identical 'B' or 'C' V-belts rated for not less than 200 % of the power (nameplate rating) of motor.
- 4.6.3 Each filter section shall have removable, 50 mm thick cleanable filters preceding the coil section.
- 4.6.4 Suitable lubrication facilities shall be provided for all fan and motor drive bearings. Grease supply and relief fitting connections shall be extended to accessible points outside the casing or fan for ease of servicing. The type of lubrication furnished shall be selected for continuous operation and minimum frequency of required servicing. Fan shaft bearings shall be sealed, designed for grease lubrication and rated for a minimum L-10 life of 100,000 hours per AFBMA.
- 4.6.5 The packaged air-conditioning units shall be tested in accordance with ARI Standard 430 and will meet the minimum capacity requirements.
- 4.6.6 Each casing enclosing the equipment shall be constructed of minimum 1.2 mm thick galvanized steel coated with baked enamel. The casing shall be suitably designed to prevent vibration being transmitted to the equipment. Each casing shall be labeled and provided with hinged or bolted access panels to permit

easy servicing of the equipment including removal and replacement of the filters, access to fan and removal of coils.

- 4.6.7 Compressors and fans of each unit shall be provided with internal spring type or rubber-in-shear vibration isolations. Isolators shall have an actual isolation efficiency of not less than 85%.
- 4.6.8 A moisture collection pan, constructed of hot dipped galvanized steel with suitable female pipe threaded drain connections and coated for corrosion protection, shall be in the cooling coil section of the casing to collect and drain condensate.
- 4.6.9 The evaporator and condenser coils shall be made of not less than 10 mm. O.D., seamless copper tubing. Plate fins shall be aluminum, mechanically bonded to copper tubes. Coils shall be tested with compressed air under water for leaks. All tests shall be in accordance with the Safety Code for Mechanical Refrigeration, ANSI 89.1. The coils shall be tested and rated in accordance with ARI Standard 410 to certify the performance.
- 4.6.10 Bidder shall furnish and install condensate drain piping from the drain pan to the nearest floor or equipment drain collection area. Drain piping shall be trapped and properly sloped. Piping shall be in accordance with Part 4, Technical Specification M-001 piping.
- 4.6.11 When Water cooled condensers will be provided, it shall be provided with all integral supply and return piping factory piped to outside of the unit's casing for external supply and return water connections. One refrigerant operated head pressure water regulating valve shall be provided to modulate condenser water flow and maintain a minimum head pressure. Condenser water side design pressure shall not be less than 7 kg/cm² (g).
- 4.6.12 Condenser shall have be provided with a spring loaded pressure relief valve which shall be piped and vented to outdoors.

4.7 INSULATION (AS APPLICABLE)

4.7.1 Refrigerating Piping Insulation

Refrigerant piping between the evaporator and compressor and all hot gas lines shall be insulated. Insulation shall be minimum 40 mm thick and shall be of the closed cell flexible tubing type. Method of applying insulation shall be in strict conformance to the manufacturer's directions.

4.7.2 Duct Insulation

- 4.7.2.1 All cold air duct shall be insulated. Insulation shall be not less than 25 mm thick and not less than 64 kg/m³ density fiberglass board with factory applied

laminate of presized, finely woven, glass cloth and foil, having a white finish suitable for painting.

- 4.7.2.2 Sealing tape shall not be less than 76 mm width and shall be the insulation manufacturer's recommended vapor proof pressure sealing tape. Color shall match insulation facing.
- 4.7.2.3 Adhesive shall be white, vapor barrier mastic and shall be compatible with the above insulation materials as recommended by the insulation manufacturer.
- 4.7.2.4 The entire insulation system shall meet NFPA 90A Fire Hazard Classification.

4.8 WELDING

Welding shall be in accordance with the applicable requirements of AWS D9.1, except as follows:

- 4.8.1 The Bidder shall qualify a welding procedure for the lightest thickness sheet to be welded under this Contract.
- 4.8.2 A qualification weld shall be performed on the lightest thickness sheet to be welded by the welder or welding operator.
- 4.8.3 Bidder shall conduct an inspection of all welds to assure conformance to AWS D9.1 criteria.
- 4.8.4 The welding of sheets, hangers, and auxiliary steel to primary structural steel shall meet the requirements of AWS D1 .1, except the Inspectors need not be certified in accordance with the provisions of AWS QC1.

4.9 CLEANING AND PAINTING

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and shall be painted In accordance with the Manufacturer's standard practice suitable for the service and environmental conditions. The Bidder shall be referred S-420 ; S-430, for Cleaning and Painting.

4.10 INSTRUMENTATION AND CONTROLS

- 4.10.1 Each wall or roof fan shall be supplied with a local hand switch and status lights.
- 4.10.2 Each centrifugal fan shall have a locally mounted hand switch along with status lights. The control panel for the fan shall include the differential pressure indicator for the intake filters. The panel shall include the status lights for the fans along with any alarm or trip indication lights.

- 4.10.3 The controls and alarms for the ventilation and air conditioning for each room or area shall be consolidated on local control panels conveniently located in that room or area. Signals shall be available for input to the plant computer to indicate malfunction of any equipment from each of the local control panels and to indicate high temperature condition.
- 4.10.4 All control panel mounted instrumentation and control devices shall be wired to electrical terminal box.
- 4.10.5 All instrumentation and control devices shall be furnished and installed by Bidder.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 IN SERVICE TESTING AND ADJUSTMENTS

- 5.1.1 The work shall be tested and also refer for M-014, additional vibration standard code, adjusted and prior to acceptance. In service testing shall be performed by Bidder in accordance requirements of these specifications.
- 5.1.2 Any deficiencies brought out during testing and balancing shall be promptly remedied by Bidder at without additional cost to the Owner.
- 5.1.3 Bidder shall provide experienced personnel, including all crafts, to assist for whatever periods required by job conditions, in the preliminary testing and starting of all operating equipment installed by Bidder.
- 5.1.4 Throughout erection and during testing, the equipment will be carefully inspected by Owner. The Bidder's general superintendent, together with such additional men as may be required, shall remain at the site of the Work, until the equipment has been placed in complete operation by Owner under the direction of Bidder who, before doing so, will thoroughly check all details. Any modifications, changes, adjustments or corrections, which may be necessary to meet the requirements under the contract, shall be done by Bidder. This corrective work shall relate only to corrections of erection defects based upon the original design of the equipment and shall be done without additional cost to Owner.

5.2 TESTING, ADJUSTING AND BALANCING

- 5.2.1 Procedures

The work shall be tested, adjusted, and balanced in accordance with qualified procedures.
- 5.2.2 System Performance Deviation Report

- 5.2.2.1 Whenever any of the following conditions occur after Bidder has achieved the optimum system balance, a system Performance Deviation Report shall be completed by Bidder. The report shall be completed within three (3) working days from whenever the unacceptable system performance values are obtained
- Airflow rates, corrected from actual test conditions to design conditions, are less than design.
 - Fan rated pressure (fan static pressure or total pressure) is greater than design and airflow rate is less than design.
 - Calculated brake power is greater than the nameplate (nominal) power rating.
 - Velocity through individual filter. Elements are 25% greater than the design values listed on manufacturer's data or drawings.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

6.1 Quality Assurance/Quality Control shall conform to be requirements of Part 4 Subsection 2.7.

6.2 The Quality control documentation as specified in Part 4 Subsection 2.7 shall include, but not be limited to, the following Shop Test Reports

- Welding Procedure
- Cleaning and Painting
- All other documentation required by Applicable Codes and Standards.
- Test and inspection reports
- Balancing report

7. ELECTRICAL AND CONTROL REQUIREMENTS

7.1 SCOPE

The Bidder shall design, furnish, install, test, commission and guarantees all electrical and control equipment required for a fully functional and reliable ventilation and air Conditioning system for the areas as described in this Specifications.

7.2 The ventilation and air conditioning (VAC) electrical and control equipment and installation shall comply with the requirements of, Part 4, Technical Specification E-001, E-208, E-210, F-413, M-001, Section 4 and as specified herein.

7.3 All ventilating fans shall be provided with locally mounted control stations. Each control station shall have an on-off switch and indicating lights for fan operating status and clogged or dirty filter. The fan failure and clogged filter shall be annunciated at the control panel.

7.4 Enclosure for local control panel and stations shall meet the requirements of IP55 for outdoor application.

(Last page of M-421)